ALHASSAN JAWAD

ELECTRICAL ENGINEER • FOCUS: CYBER- & OT-SECURITY

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ABOUT ME

A newly graduated **Electrical Engineer** (B.Sc.) with 2+ years of academic experience as a Teaching Assistant, focusing on technical evaluation and laboratory verification. During my studies, I focused on control theory and the programming of embedded systems. Currently in the final phase of getting my M.Sc. (Graduation: Q1 2027) in electrical engineering focusing on **OT-security** and **control systems**.

My background bridges electrical infrastructure with embedded systems, ranging from low-level C programming to software development. I apply **control theory**, **signal processing**, and **IoT frameworks** to engineer secure software solutions. My goal is to join a collaborative environment where I can continuously develop my skills, adapt to new challenges, and make a meaningful impact on **critical infrastructure**.

TECHNICAL COMPETENCIES

Programming Languages: Python, Java, C/C++, JavaScript, HTML, CSS, MATLAB, LabVIEW

Engineering Tools & Frameworks: Angular, Tailwind CSS, npm, Altium Designer, PSpice, LTSpice, PostgreSQL

Engineering Skills: Automation, Control Systems, Embedded Systems, Signal Processing, Data Analysis

Languages: Swedish (Fluent), English (Fluent), Arabic (Native)

RELEVANT EXPERIENCE

Software Engineer (Internship) – Portabel Health
Sep 2025 – Dec 2025

Programmed and debugged system-level components for the production platform using:

- Angular framework,
- Tailwind CSS,
- Node Package Manager (npm)

Executed over 130 technical tasks within an Agile development pipeline using Freedcamp, focusing on feature deployment and continuous testing across Dev and QA environments.

Managed backend data structures using PostgreSQL and translated UI/UX designs from Figma into functional code. To ensure high availability and prevent system downtime during updates, I wrote comprehensive script refactoring documentation for each new script, and created a standardized internal Git version control guideline.

Teaching Assistant – Uppsala University
Aug 2023 – Jul 2025

- Instructed and evaluated 50+ students per semester on core programming & electrical concepts.
- Was responsible for grading parts of each year's assignments and final exam.
- Collaborated with faculty and fellow peers to develop new course materials.

Co-founder & Chief Technology Officer (CTO) – NordicFlip / Furrytool
Mar 2023 – Jun 2024

Startups (Closed down)

Co-founded and engineered both startups from initial conceptual design to a functional small business platform, scaling the system to support approximately 100 global buyers prior to operations closing. Managed the full-stack technical lifecycle, executing all development tasks from initial system architecture design down to code-level implementation and platform deployment.

EDUCATION

Master of Science (M.Sc.) in Electrical Engineering 2025 – 2027

Uppsala University – Specialization: Control of Electrical Systems & Programming of Embedded Systems

Bachelor of Science (B.Sc.) in Electrical Engineering 2019 – 2022

Department of Electrical Engineering | Uppsala University
Degree issued in 2026

SELECTED PROJECTS

Master's Thesis: Automated Aeroponic Cultivation System 2025

Internal Collaboration with Uppsala University

- Programmed and deployed a rule-based MIMO (Multiple-Input Multiple-Output) control loop on an ESP32 microcontroller to measure and regulate internal temperature and humidity values.
- Integrated a real-time IoT sensor network to establish a continuous data acquisition loop, thus verifying closed-loop feedback stability and firmware execution.
- Compiled technical documentation for system initialization, debugging, and multi-variable tuning routines.

Bachelor's Thesis: Smart Building Energy Management 2022

External Collaboration with Studenternas IP, Uppsala

- Wrote a custom optimization script in MATLAB to manage energy consumption and load distribution for a commercial property.
- Processed multi-variable datasets from deployed IoT sensors, isolating 71.2 kW of the facility's 185 kW total load as manageable power flexibility.
- The system verified that 38.5% of the infrastructure's peak demand could be dynamically controlled and deferred without affecting facility operations.

Project: Miniature Elevator Prototype (Course: Project in Electronics) 2023

Academic Project | Uppsala University

- Helped in designing a three-level miniature elevator, handling end-to-end HIL (Hardware-In-the-Loop) integration.
- Programmed the control logic and GUI (in LabVIEW), to manage real-time level selection and state transitions while regulating motor speed profiles, to achieve a smooth acceleration while minimizing physical oscillations.

Project: GPS Tracking Device (Course: PC-Board Construction with ECAD Tools) 2022

Academic Project | Uppsala University

- Designed a multi-layer GPS tracking Electronic PCB within Altium Designer.
- Integrated an ATmega328P microcontroller with a Quectel MC60 transceiver to process satellite GPS data and transmit coordinates over different networks.
- Implemented a hardware power-saving scheme by interfacing an MMA8653 accelerometer to provide system wake-up triggers based on physical motion.